

WEEK 7 REPORT

Student Performance - Data Cleaning & Transformation Using Power BI

1. Introduction:

During Week 7, I worked on the **Student Performance in Exams** dataset using Microsoft Power BI. The main focus of this activity was to clean, transform, and prepare the dataset for further analysis.

I used **Power Query Editor** to inspect the dataset, check data types, handle data inconsistencies, create calculated columns, classify student performance, and reshape the subject-wise score data. I also created basic DAX measures for analyzing student performance.

The overall process helped me understand how raw data can be converted into a structured and analysis-ready dataset using Power BI.

2. Objective:

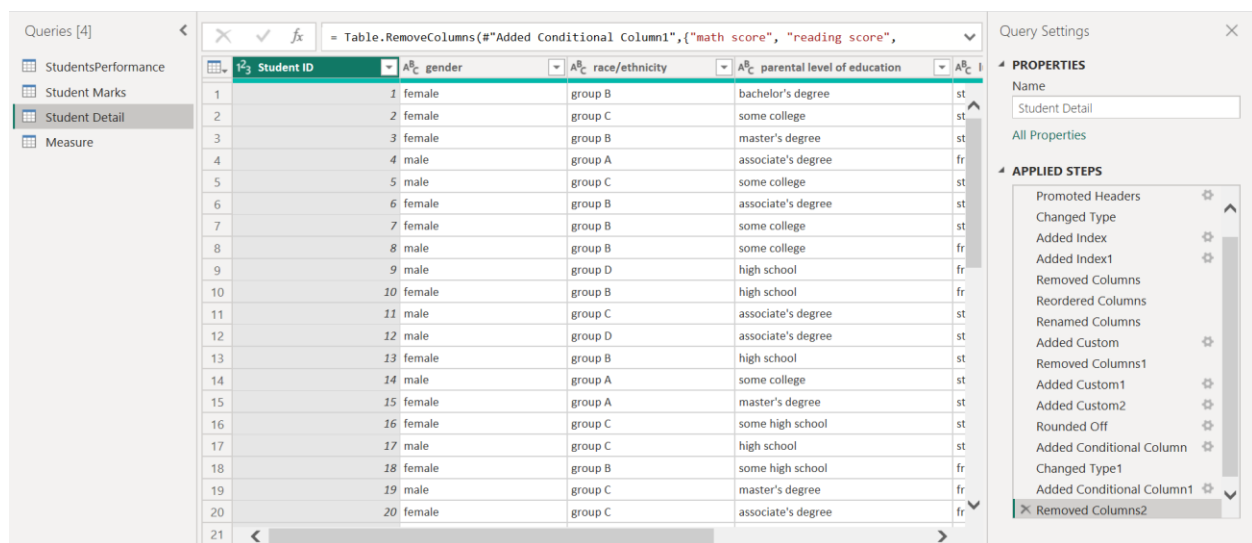
The main objectives of this week's activity were:

- To import the Student Performance in Exams dataset into Power BI.
- To inspect the rows, columns, and data types.
- To identify missing, duplicate, or invalid data.
- To clean and standardize the dataset.
- To transform categorical attributes.
- To create Total Score and Average Score columns.
- To create Performance Grade categories.
- To classify students as Pass or Fail.
- To reshape subject-wise data using Unpivot.
- To create basic DAX measures.
- To validate the transformed dataset for further analysis.

3. Importing the Dataset:

I started by opening **Power BI Desktop** and selecting **Get Data**. I imported the Student Performance in Exams dataset using the appropriate file format.

After selecting the dataset, I chose **Transform Data** to open the **Power Query Editor**. Power Query Editor was used to perform the required data cleaning and transformation operations.



The screenshot displays the Power Query Editor interface. On the left, the 'Queries' pane shows 'Student Detail' selected. The main area shows a table with columns: Student ID, gender, race/ethnicity, and parental level of education. The table contains 20 rows of data. On the right, the 'Query Settings' pane shows the 'APPLIED STEPS' list, which includes: Promoted Headers, Changed Type, Added Index, Added Index1, Removed Columns, Reordered Columns, Renamed Columns, Added Custom, Removed Columns1, Added Custom1, Added Custom2, Rounded Off, Added Conditional Column, Changed Type1, Added Conditional Column1, and Removed Columns2.

Student ID	gender	race/ethnicity	parental level of education
1	female	group B	bachelor's degree
2	female	group C	some college
3	female	group B	master's degree
4	male	group A	associate's degree
5	male	group C	some college
6	female	group B	associate's degree
7	female	group B	some college
8	male	group B	some college
9	male	group D	high school
10	female	group B	high school
11	male	group C	associate's degree
12	male	group D	associate's degree
13	female	group B	high school
14	male	group A	some college
15	female	group A	master's degree
16	female	group C	some high school
17	male	group C	high school
18	female	group B	some high school
19	male	group C	master's degree
20	female	group C	associate's degree

4. Exploring and Inspecting the Dataset:

After importing the dataset, I examined the available rows and columns to understand the structure of the data.

The dataset contained both **numerical and categorical attributes**. The main student-related categorical attributes included:

- Gender
- Race/Ethnicity
- Parental Education
- Lunch
- Test Preparation Course

The numerical attributes mainly consisted of:

- Math Score
- Reading Score
- Writing Score

I also checked the data types of the columns and inspected the dataset for missing values, errors, and duplicate records.

5. Cleaning the Dataset:

The next step was to clean the imported dataset. Data cleaning was performed to make sure that the data was suitable for calculations and analysis.

The following checks and transformations were performed:

- Checked for duplicate records.
- Checked for missing or invalid values.
- Corrected inappropriate data types where necessary.
- Reviewed categorical values for consistency.
- Renamed columns where required for better understanding.

Data cleaning is important because incorrect or inconsistent data can affect the accuracy of the analysis.

Queries [4]

StudentsPerformance

Student Marks

Student Detail

Measure

Table: RemoveColumns(#"Added Conditional Column1",{"math score", "reading score", "writing score"})

	race/ethnicity	parental level of education	lunch	test preparation course
1	usp B	bachelor's degree	standard	none
2	usp C	some college	standard	completed
3	usp B	master's degree	standard	none
4	usp A	associate's degree	free/reduced	none
5	usp C	some college	standard	none
6	usp B	associate's degree	standard	none
7	usp B	some college	standard	completed
8	usp B	some college	free/reduced	none
9	usp D	high school	free/reduced	completed
10	usp B	high school	free/reduced	none
11	usp C	associate's degree	standard	none
12	usp D	associate's degree	standard	none
13	usp B	high school	standard	none
14	usp A	some college	standard	completed
15	usp A	master's degree	standard	none
16	usp C	some high school	standard	none
17	usp C	high school	standard	none
18	usp B	some high school	free/reduced	none
19	usp C	master's degree	free/reduced	completed
20	usp C	associate's degree	free/reduced	none

Query Settings

PROPERTIES

Name

Student Detail

APPLIED STEPS

Promoted Headers

Changed Type

Added Index

Added Index1

Removed Columns

Reordered Columns

Renamed Columns

Added Custom

Removed Columns1

Added Custom1

Added Custom2

Rounded Off

Added Conditional Column

Changed Type1

Added Conditional Column1

Removed Columns2

6. Transforming Categorical Attributes:

I reviewed the important categorical columns in the dataset and checked whether their values were consistent.

The following attributes were considered:

- Gender
- Race/Ethnicity
- Parental Education
- Lunch
- Test Preparation Course

The categorical values were standardized wherever required so that they could be used correctly during analysis and visualization.

7. Creating Calculated Columns:

I created new calculated columns using the three subject scores available in the dataset.

The subject scores were:

- Math Score
- Reading Score
- Writing Score

Total Score

The **Total Score** was calculated by adding the marks obtained in all three subjects.

Total Score = Math Score + Reading Score + Writing Score

This column represents the student's total marks across the three subjects.

Average Score

The **Average Score** was calculated using the total score.

$$\text{Average Score} = \text{Total Score} / 3$$

The Average Score helps to understand the overall academic performance of each student.

The image displays two screenshots of the 'Custom Column' dialog box in Power Query, illustrating the steps to create a new calculated column.

First Screenshot:

- Title:** Custom Column
- Instruction:** Add a column that is computed from the other columns.
- New column name:** Total Marks
- Custom column formula:** = [math score]+[reading score]+[writing score]
- Available columns:** A list of columns including 'lunch', 'test preparation course', 'math score', 'reading score', 'writing score', 'Total Marks', and 'Average'. The 'writing score' column is currently selected.
- Buttons:** '<< Insert', 'OK', and 'Cancel'.
- Status:** No syntax errors have been detected.

Second Screenshot:

- Title:** Custom Column
- Instruction:** Add a column that is computed from the other columns.
- New column name:** Average
- Custom column formula:** = [Total Marks]/3
- Available columns:** The same list of columns as the first screenshot, but now 'Total Marks' is selected.
- Buttons:** '<< Insert', 'OK', and 'Cancel'.
- Status:** No syntax errors have been detected.

8. Creating Performance Grade:

To make student performance easier to understand, I created a **Performance Grade** using conditional logic.

The performance was classified into four categories:

Grade	Performance Level
A	High Performance
B	Good Performance
C	Average Performance
F	Low Performance

The grade was created based on the student's Average Score using conditional classification.

This classification makes it easier to group students according to their overall performance.

×

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

	Column Name	Operator	Value		Output
If	Average	is greater than or...	80	Then	A
Else If	Average	is greater than or...	60	Then	B
Else If	Average	is greater than or...	40	Then	C

Else

9. Creating Pass/Fail Classification:

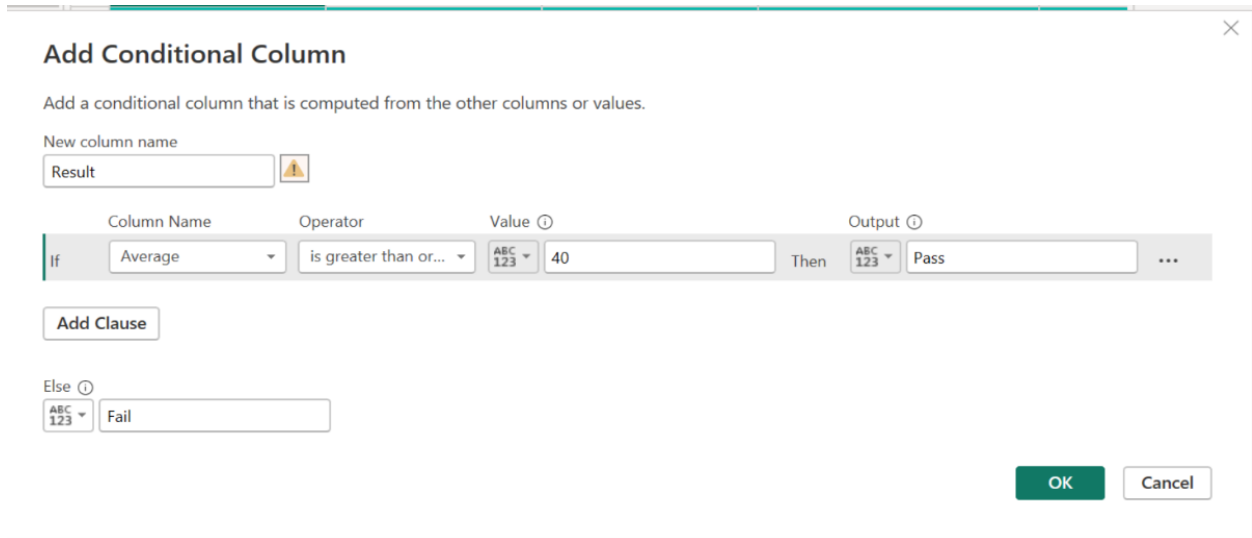
I created a **Pass/Fail** classification to identify whether a student met the selected passing criterion.

A suitable score condition was applied using a **Conditional Column**.

The resulting categories were:

- Pass
- Fail

This classification can be used to identify successful students and calculate the overall pass percentage.



Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Result

	Column Name	Operator	Value		Output
If	Average	is greater than or...	40	Then	Pass

Add Clause

Else
Fail

OK Cancel

10. Reshaping Subject Data Using Unpivot:

The original dataset contained separate columns for:

- Math Score
- Reading Score
- Writing Score

To make subject-wise analysis easier, I used the **Unpivot Columns** operation in Power Query.

After applying Unpivot, the subject-related information was reorganized into two common columns:

- Subject
- Score

This transformation allows the performance of different subjects to be analyzed using a common structure.

Queries [4] | = Table.RenameColumns(#"Unpivoted Columns",{{"Attribute", "Subject"}, {"Value", "Score"}})

Student ID	Subject	Score
1	math score	72
2	reading score	72
3	writing score	74
4	math score	69
5	reading score	90
6	writing score	88
7	math score	90
8	reading score	95
9	writing score	93
10	math score	47
11	reading score	57
12	writing score	44
13	math score	76
14	reading score	78
15	writing score	75
16	math score	71
17	reading score	83
18	writing score	78
19	math score	88
20	reading score	95
21	writing score	92

Query Settings: Student Marks

APPLIED STEPS: Added Index, Added Index1, Removed Columns, Reordered Columns, Renamed Columns, Added Custom, Removed Columns1, Added Custom1, Added Custom2, Rounded Off, Added Conditional Column, Changed Type1, Added Conditional Column1, Removed Columns2, Unpivoted Columns, **Renamed Columns1**

11. Preparing the Data Model:

After completing the transformations, I reviewed the structure of the transformed dataset.

I checked the following:

- The transformed columns were available.
- Unnecessary columns were removed where required.
- The table structure was organized properly.
- The required fields were available for analysis.
- The data was suitable for creating measures and visualizations.

The prepared structure can now be used for further Power BI analysis.

12. Loading the Transformed Data:

After completing the required transformations in Power Query Editor, I selected **Close & Apply**.

This loaded the transformed data into Power BI Desktop.

After loading the data, I verified that:

- Total Score was available.
- Average Score was available.
- Performance Grade was available.
- Pass/Fail classification was available.
- The unpivoted subject data was correctly loaded.

13. Creating DAX Measures:

After loading the transformed dataset, I created basic DAX measures to support student performance analysis.

The measures included:

Average Math Score

Average Math Score = AVERAGE('Student Performance'[Math Score])

Average Reading Score

Average Reading Score = AVERAGE('Student Performance'[Reading Score])

Average Writing Score

Average Writing Score = AVERAGE('Student Performance'[Writing Score])

Overall Average Score

Overall Average Score = AVERAGE('Student Performance'[Average Score])

Overall Pass %

Overall Pass % =
DIVIDE(

```

CALCULATE(COUNTROWS('Student Performance'), 'Student
Performance'[Pass/Fail] = "Pass"),
COUNTROWS('Student Performance')
) * 100

```

These measures provide summarized values that can be used in Power BI visualizations and dashboards.

General										
General										
From Number										
From Date & Time										
All Insights										
Queries [3]										
Students/Performance										
Student Marks										
Student details										
Table.RemoveColumns(#"Added Conditional Column1",{"math score", "reading score", "writing score", "Total Marks", "Average", "Grade",										
Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course					
1	Female	Group B	Bachelor's Degree	Standard	None	Valid	100%	Valid	100%	Valid
2	Female	Group C	Some College	Standard	Completed	Error	0%	Error	0%	Error
3	Female	Group B	Master's Degree	Standard	None	Empty	0%	Empty	0%	Empty
4	Male	Group A	Associate's Degree	Free/Reduced	None	Valid	100%	Valid	100%	Valid
5	Male	Group C	Some College	Standard	None	Error	0%	Error	0%	Error
6	Female	Group B	Associate's Degree	Standard	None	Empty	0%	Empty	0%	Empty
7	Female	Group B	Some College	Standard	Completed	Valid	100%	Valid	100%	Valid
8	Male	Group B	Some College	Free/Reduced	None	Error	0%	Error	0%	Error
9	Male	Group D	High School	Free/Reduced	Completed	Empty	0%	Empty	0%	Empty
10	Female	Group B	High School	Free/Reduced	None	Valid	100%	Valid	100%	Valid
11	Male	Group C	Associate's Degree	Standard	None	Error	0%	Error	0%	Error
12	Male	Group D	Associate's Degree	Standard	None	Empty	0%	Empty	0%	Empty
13	Female	Group B	High School	Standard	None	Valid	100%	Valid	100%	Valid
14	Male	Group A	Some College	Standard	Completed	Error	0%	Error	0%	Error
15	Female	Group A	Master's Degree	Standard	None	Empty	0%	Empty	0%	Empty
16	Female	Group C	Some High School	Standard	None	Valid	100%	Valid	100%	Valid
17	Male	Group C	High School	Standard	None	Error	0%	Error	0%	Error
18	Female	Group B	Some High School	Free/Reduced	None	Empty	0%	Empty	0%	Empty
19	Male	Group C	Master's Degree	Free/Reduced	Completed	Valid	100%	Valid	100%	Valid
20	Female	Group C	Associate's Degree	Free/Reduced	None	Error	0%	Error	0%	Error
21	Male	Group D	High School	Standard	None	Empty	0%	Empty	0%	Empty
22	Female	Group B	Some College	Free/Reduced	Completed	Valid	100%	Valid	100%	Valid
23	Male	Group D	Some College	Standard	None	Error	0%	Error	0%	Error

14. Validating the Results:

After completing the cleaning and transformation process, I validated the results to make sure that the transformed data was correct.

I checked whether:

- The Total Score was calculated correctly.
- The Average Score was calculated correctly.
- Performance Grade classifications were working properly.
- Pass/Fail classifications were correctly assigned.
- The unpivoted subject data contained the required values.
- The DAX measures produced appropriate results.
- The final dataset was ready for visualization.

The validation process helped ensure that the transformed dataset was reliable for further analysis.

15. Week 7 Output:

At the end of Week 7, I successfully prepared a **cleaned, transformed, and structured Student Performance dataset**.

The final dataset included:

- Cleaned student information
- Subject-wise scores
- Total Score
- Average Score
- Performance Grade
- Pass/Fail classification
- Unpivoted Subject and Score fields
- Basic DAX measures

The prepared dataset is now suitable for further analysis and dashboard development in Power BI.

Average math score
66.09

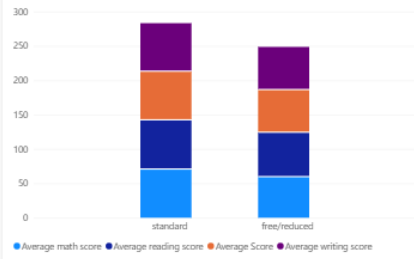
Average Score
67.77

Overall Pass %
0.97

Average reading score
69.17

Average writing score
68.05

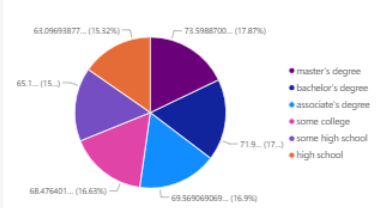
Average math score, Average reading score, Average Score and Average writing score by lunch



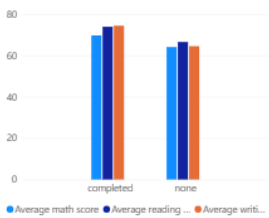
Average math score, Average reading score and Average writing score by race/ethnicity



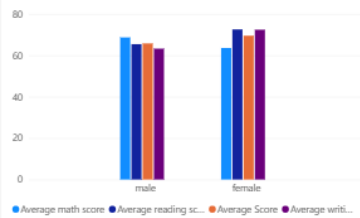
Average Score by parental level of education



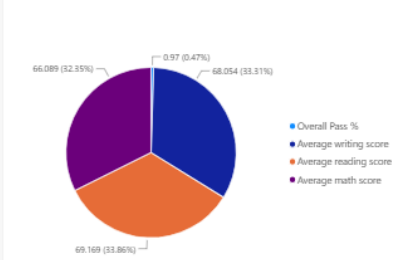
Average math score, Average reading score and Average writing score by test preparation course



Average math score, Average reading score, Average Score and Average writing score by gender



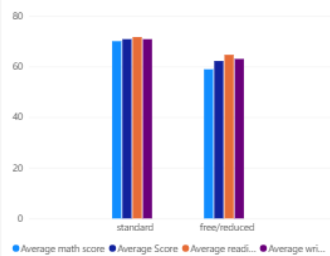
Overall Pass %, Average writing score, Average reading score and Average math score



race/ethnicity, parental level of education, lunch

- ☐ group A
- ☐ group B
- ☐ group C
- ☐ group D
- ☐ group E

Average math score, Average Score, Average reading score and Average writing score by lunch



16. Conclusion:

Week 7 focused on cleaning and transforming the Student Performance in Exams dataset using Microsoft Power BI.

I completed the process of importing and inspecting the data, cleaning inconsistencies, transforming categorical attributes, creating Total Score and Average Score, classifying students using Performance Grade and Pass/Fail categories, reshaping subject data using Unpivot, and creating basic DAX measures.

The final result was a structured and analysis-ready dataset that can be used for the next stage of Power BI analysis and dashboard development.

Submitted By:

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